

REWEAR-FUSED is a high-ceiling, genuinely novel concept with strong shock factor for the Carter's "Making & Remaking" track, but it has multiple severe, addressable vulnerabilities that could sink it in judging or execution. It is not killed by existing prior art as of June 1 2026, yet the enzyme-design and polymer-design fields are moving extremely fast. The 72-hour constraint amplifies every technical risk. Below is a decision-grade red-team analysis based on exhaustive searches across arXiv, bioRxiv, ChemRxiv, Science/Nature portfolios, JACS, ACS Catalysis, Angewandte, Green Chemistry, patents, company disclosures, and recent preprints (through late May 2026).

1. Prior-Art Kill-Shots (or Lack Thereof)

No direct kill-shot exists for either novelty flag, but the margin is narrower than claimed and shrinking rapidly.

De novo carbamate/urethane hydrolase:

No publicly disclosed de novo (backbone + active-site) computationally designed carbamate hydrolase/urethanase exists. UMG-SP2 (and homologs UMG-SP1/3) are metagenomic discoveries (Branson et al. 2023 and follow-ups); they use the Ser-Ser^{cis}-Lys triad characteristic of the amidase signature superfamily. Paiva et al. (Chem. Sci. 2025, 16, 2437–2452; first published online 18 Dec 2024, DOI:10.1039/d4sc06688j) and Świderek et al. (JACS 2025, DOI:10.1021/jacs.5c13147) provide QM/MM mechanistic detail on UMG-SP2: two-stage mechanism with acylation (ester-like cleavage of the urethane bond, $\Delta G^\ddagger \approx 20.8$ kcal/mol, rate-limiting) releasing alcohol, followed by deacylation to unstable carbamic acid $\rightarrow$ amine + CO₂. Engineering has been limited to site-directed/semi-rational mutagenesis (e.g., Li et al. Adv. Sci. 2025, variants A141G/Q399A with >30-fold gains on polyester-PUR).

Recent de novo serine hydrolase work is the closest threat: Lauko, Pellock, Sumida, Baker et al., *Science* **388**, eadu2454 (2025; bioRxiv 29 Aug 2024, DOI:10.1126/science.adu2454). They used RFdiffusion to scaffold complex active sites (including multi-step serine hydrolase mechanisms) + PLACER/ChemNet ensemble analysis for preorganization across the reaction coordinate, producing functional esterases ($k_{\text{cat}}/K_{\text{M}}$ up to $\sim 3.8 \times 10^3 \text{ M}^{-1} \text{ s}^{-1}$) with novel folds and high design-model agreement (C α RMSD < 1 Å). The method explicitly supports specifying catalytic residue geometries (Ser-based triads) around a substrate. No carbamate/urethane substrate was used, but the pipeline is directly transferable to a Ser-Ser^{cis}-Lys theozyme anchored on the UMG-SP2 transition state. Boltz-2 (Passaro, Wohlwend et al., bioRxiv 14 Jun 2025 / MIT-Recursion 2025; extends AF3 with binding-affinity prediction) further strengthens in-silico validation.

Likelihood of a scoop before 17 Jun 2026: Moderate-to-high for a general serine hydrolase on a carbamate model substrate from the Baker lab or competitors (e.g., extensions of Kim/Ahern metallohydrolase work, *Nature* 2025). Exact match to elastane-relevant high-MW PU is unlikely in the next 16 days, but a preprint applying the Lauko pipeline to amidase-signature geometry would erode the "first de novo urethanase" claim significantly.

AI-generative cleavable elastane/segmented PU:

No published generative design of an enzymatically cleavable segmented polyurethane elastane/spandex variant exists. POLYT5 (Sahu, Ramprasad et al., *npj Artif. Intell.* **2**, 30, 2026; DOI:10.1038/s44387-026-00087-1, announced ~24 Mar 2026) is the strongest recent advance: encoder-decoder T5-style chemical language model trained on >12k experimental polymers + 100M hypotheticals, including polyurethanes. It generates chemically valid structures conditioned on properties and is explicitly noted as extensible to block/random copolymers via fine-tuning on curated datasets (with tokens for composition/architecture). polyBART (Savit et al., EMNLP Findings 2025) is the bidirectional predecessor. Vitrimer work (graph VAE + MD inverse design for transesterification vitrimers with targeted T_g, arXiv 2023–2024) shows generative models can target recyclability, but not enzymatic carbamate cleavage in phase-separated segmented PUs. No joint polymer-enzyme co-design loop (polymer emitting substrate features → enzyme constraint feeding back as fiber accessibility/sterics) has been demonstrated for any textile fiber.

Joint co-design: Zero hits. Closest are separate eco-design efforts (e.g., Aarsen et al., *Chem. Rev.* 2024 on tailoring polyesters for circularity via hydrolyzable bonds) or enzyme-polymer interaction simulations, but nothing closed-loop.

Verdict on novelty flags: Both remain open as of 1 Jun 2026. Closest scoop risk is Baker-group serine hydrolase extensions (weeks-to-months horizon) and Ramprasad-group POLYT5 fine-tunes targeting sustainable elastomers (already validated in lab for other properties). No one has done the *joint* loop or the exact carbamate-anchored de novo urethanase + cleavable segmented PU pair. Your narrative (“neither existed 72 hours ago”) holds for the hackathon window.

2. Technical Feasibility Holes (72-Hour Reality Check)

This will break in multiple places; the single biggest failure mode is **enzyme inaccessibility on real elastane fiber due to hard-segment crystallinity**, not compute or model architecture.

- **Segmented-copolymer extension of polyBART/POLYT5:** Tractable but high-risk in 72 h. POLYT5 already includes polyurethanes in training and supports block-copolymer fine-tuning via additional tokens for hard/soft ratios and repeat-unit sequences. Specifying “carbamate linkages sterically accessible” requires either (a) explicit cleavage-site tokens + property conditioning (T_g soft < -50 °C, hard > 150 °C or T_m, elongation ≥ 400 %) or (b) post-generation MD screening for phase separation and surface-exposed urethane bonds. Neither is plug-and-play; you will need custom tokenization, fine-tuning data curation, and integration with polymer property predictors or coarse-grained MD. Feasible for a strong team with prior polymer-ML experience, but a dead-end if you treat it as “just prompt the model.” Vitrimer VAE precedents show inverse design works for dynamic bonds; segmented PU adds morphology prediction burden.

- **Theozyme specification for Ser–Ser^{cis}–Lys carbamate mechanism:** Sound in principle (exact match to UMG-SP2 QM/MM transition state from Paiva/Świderek 2025). RFDiffusion2/3 + LigandMPNN can scaffold it; the Lauko 2025 pipeline already succeeded for multi-step serine mechanisms. However, carbamate substrates are larger and more polar than the 4MU esters used in Lauko; theozyme placement must account for the –O–C(=O)–NH– geometry and leaving-group stabilization. PLACER/ChemNet-style ensemble validation across acylation/deacylation steps is essential and already demonstrated.
- **Boltz-2 + PLACER validation credibility for carbamate substrate:** High for *in silico* proof-of-concept. Boltz-2 (2025) jointly predicts structure + affinity and handles ligands; it is a major upgrade over prior tools for designed enzyme–substrate complexes. Combined with PLACER preorganization scoring, you can generate credible “this design should work” figures. Limitation: Boltz-2 affinity is tuned more for small-molecule drugs than polymeric substrates; fiber docking/MD will be needed for realism. Still, far better than nothing for a 72 h demo.
- **GPU/compute realities:** RFDiffusion inference, LigandMPNN, Boltz-2, and polymer model sampling are GPU-heavy but doable. Dozens of designs + validation ensembles are feasible on 4–8× A100/H100 in a university cluster or rented cloud over 72 h (many hackathon teams do this). The co-design loop (iterative: generate polymer → extract target carbamate microenvironment → design enzyme → score accessibility → constrain next polymer round) is the killer—each iteration could take hours. You will almost certainly run a single-pass or very shallow loop, not true closed-loop optimization.

Single biggest technical failure mode: Hard-segment crystallinity in any realistic segmented PU (even your designed variant) creates physical barriers to enzyme diffusion and active-site access. Multiple studies confirm crystalline urethane/urea domains (H-bonded) severely limit enzymatic attack on high-MW elastane fibers; degradation is mostly surface-limited or requires pretreatment/swelling (see Azevedo et al. 2025 on DMSO/DBN elastane removal; general reviews on PU biodegradation 2025). Your “sterically accessible carbamate linkages” premise is optimistic unless the generative model explicitly penalizes excessive hard-segment crystallinity or places cleavable sites preferentially in amorphous/soft regions while preserving mechanics. This is the #1 reason the wet-lab validation path for the residency could fail.

3. Scientific Soundness

- **Mechanical-property targets:** Reasonable and aligned with commercial spandex (elongation 400–800 %, soft-block T_g typically –50 to –70 °C for PTMEG polyethers, hard-block T_g/T_m high for strength via H-bonding/ π -stacking). Achievable in segmented PUs via soft-segment molecular weight and hard-segment content (Yilgör et al. 2015 classic review). The generative model can target these via conditioning; the risk is that adding “cleavable” features (e.g., sterically exposed carbamates or disrupted H-bonding) trades off elasticity/recovery.
- **Carbamate-accessibility premise:** Partially sound but high-risk. UMG-SP2 and engineered variants do cleave urethane bonds in *low-MW dicarbamates and some*

thermoplastic PUs, but activity on actual elastane *fibers* (high crystallinity, high MW, phase-separated) is modest or unproven at scale. Crystallinity is the dominant barrier (crystalline lamellae restrict water/enzyme diffusion; see 2025 reviews on enzymatic PU degradation). Designing “sterically accessible” linkages helps, but if hard segments remain crystalline to deliver the required mechanics, enzymes will still struggle. Soundness improves if you explicitly optimize for reduced hard-segment crystallinity or surface-enriched cleavable sites.

- **Anchoring on UMG-SP2 Ser–Ser^{cis}–Lys transition state:** Excellent choice. It is the native, validated geometry for carbamate cleavage (not a generic serine hydrolase). The Lauko pipeline already handles complex, multi-step geometries; you are not starting from zero.
- **Co-design loop coherence:** Scientifically coherent in concept (substrate features from polymer → enzyme target; enzymatic success metric → polymer constraint on accessibility/sterics). Not hand-wavy at the high level, but implementation details (quantitative feedback function, convergence criteria, handling of polymer morphology) are underdeveloped in the proposal. It is a genuine research direction, not just marketing.

Overall soundness: High on paper, but the crystallinity-accessibility tension is the soft underbelly that wet-lab validation will expose first.

4. Higher-Ceiling Alternatives (or Upgrades to REWEAR-FUSED)

REWEAR-FUSED is already extremely audacious. Nothing clearly beats it on the exact battlefield (children’s apparel circularity + enzyme-closed loop + Carter’s sponsorship), but targeted upgrades raise the ceiling without diluting the core narrative:

- **Enzyme cocktail / multi-enzyme system (strong upgrade):** Design two *de novo* enzymes in parallel (urethanase for hard-segment carbamates + esterase or amidase for soft-segment linkages or residual bonds). Many PU recycling papers emphasize that single enzymes are insufficient; cocktails or consortia achieve deeper depolymerization (Wei lab and others, 2025). Narrative upgrade: “one enzyme for the stretch, one for the rest—full textile-to-monomer in one pot.” Higher technical ceiling, still fits 72 h if you parallelize (same pipeline, different theozymes). Beats pure single-enzyme version on completeness.
- **Vitrimer-like dynamic bonds + enzymatic trigger (hybrid, very high ceiling):** Incorporate reversible/dynamic linkages (e.g., the carboxylate or disulfide vitrimer chemistries already demonstrated in segmented PUs) that the generative model can place, then design an enzyme that *catalyzes* exchange or cleavage under mild conditions. Combines chemical recyclability (reprocessible without full depolymerization) with enzymatic “kill switch.” More defensible for scale-up than pure enzymatic fiber breakdown. Precedents exist (dynamic segmented PU vitrimers 2020–2021); generative extension via POLYT5 fine-tune is plausible.

- **PFAS or other irrelevant detours:** No—children’s apparel has PFAS scrutiny (California AB 1817 etc.), but standard spandex does not contain them. Designing defluorinases adds zero value here.
- **Engineered microbial consortia or embedded catalysts:** Too far for 72 h and weakens the “clean molecule + clean enzyme” pitch.

Recommendation: Keep REWEAR-FUSED core but upgrade to **dual-enzyme co-design** (urethanase + complementary hydrolase) + explicit crystallinity penalty in the polymer objective. This increases shock factor and addresses the #1 failure mode. If you must pivot, a “designed-to-degrade” segmented PU with *both* enzymatic and mild-chemical (e.g., base-catalyzed) triggers would be even more commercially robust for Carter’s, but loses some of the pure “new enzyme” novelty.

5. Prize-Strategy Critique (Carter’s + Cox Cleantech Residency Lens)

What weakens the pitch: Demoting the compliance layer to “~10% minor” is a mistake. Carter’s makes *children’s/baby apparel*—CPSIA, California AB 1817/347 (PFAS/textile chemicals), NY ECL, Maine LD 1503, EU ESPR/Digital Product Passport, and OEKO-TEX are *existential* for any new fiber/enzyme system entering their supply chain. Judges (especially industry-side) will immediately ask: “What monomers or degradation products result? Are they safe for skin contact/ingestion in infants? Has this been screened against the CPSC Recalls or EU Safety Gate databases?” Training a lightweight router on public data is table stakes, not optional. Make it a visible, validated component (even if rule-based + ML classifier) or you lose on deployability.

Strongest hostile judge questions (and answers):

- “Your designs are purely in silico—zero wet-lab data. Why should we believe they will work on real garments or survive industrial dyeing/finishing?” → “Exactly why the \$10K + 6-week residency exists: to synthesize the top 3–5 polymer candidates, express/purify the enzyme, and run fiber-spinning + enzymatic depolymerization assays on prototype stretch fabrics. We have already de-risked the hardest part (the joint design loop) computationally.”
- “Existing urethanases + mechanical/chemical recycling already exist—why invent new ones?” → “Because current enzymes have negligible activity on crystalline elastane fibers in real apparel blends, and chemical routes use toxic solvents (DMF/DMAc) banned or restricted for kids’ products. Our co-designed pair is the first to target *both* the molecule *and* its specific cleavage enzyme simultaneously.”
- “How does this scale to Carter’s millions of units?” → Residency deliverable includes techno-economic analysis and supply-chain integration roadmap (enzyme production via fermentation, fiber compatibility with existing spandex lines).

Most likely reason you lose: Over-ambition without a crisp, visual, *demo-able* artifact in 72 h (e.g., a compelling 3D-rendered co-design dashboard + in-silico validation movie + clear “before/after” fiber property plots) combined with a weaker compliance/safety story. Teams that

deliver a narrower but fully simulated + visualized prototype + explicit regulatory path will beat pure ambition. The grand-prize bar (World Cup tickets) favors something that feels “impossible yet almost real.”

Ranked Threats (Most to Least Severe) + Specific Fixes

1. **Enzyme inaccessibility on crystalline hard segments (highest severity – scientific + execution killer).** Fix: Add explicit crystallinity penalty or amorphous-region enrichment objective to the polymer generative loss; run short MD on top designs to quantify surface-exposed carbamate fraction; target dual-enzyme system (one optimized for hard-segment geometry).
2. **72-hour execution of the full co-design loop (or even single high-quality designs + validation).** Fix: Pre-stage tokenization/fine-tuning scripts and theozyme templates before kickoff (June 14). Run shallow single-pass (polymer → enzyme) rather than iterative loop; prioritize 5–10 top designs with Boltz-2 + PLACER scoring and clear visuals. Use Georgia Tech POLYT5 as base (local proximity advantage?).
3. **Compliance/safety narrative weakness.** Fix: Elevate to first-class component—build and demo a simple classifier/router trained on the cited public datasets; include explicit discussion of degradation products (amines, CO₂, recovered polyols) and safety screening plan for residency.
4. **Prior-art erosion (moderate, time-sensitive).** Fix: Monitor Baker lab / Ramprasad lab preprints daily; be ready to differentiate (“first *joint* co-design + first *carbamate-specific* de novo anchoring on UMG-SP2 geometry”).
5. **Mechanical targets vs. cleavability trade-off.** Fix: Validate generated polymers against known spandex property datasets; include recovery/elasticity metrics in conditioning.

Final Verdict: REWEAR-FUSED is the best possible concept for this hackathon if you execute the fixes above—particularly the crystallinity/accessibility mitigation and elevated compliance layer. It delivers the exact “shock factor + deep-tech residency fit + Carter’s circularity narrative” the sponsors want, with open novelty flags and a defensible path to the \$10K/6-week validation. It is *not* “safe/feasible-for-small-team”; that is its strength for maximal ambition. Do not pivot to something tamer. Upgrade to dual-enzyme + explicit morphology control and you maximize both track win probability and grand-prize upside. The 72-hour limit makes it a moonshot, but moonshots win these prizes when the science is this coherent and the visuals are compelling. Go execute the hard version—you have the right target.

Sources

Thinking about your request

Searching for prior-art urethanases

Searched web

de novo urethanase OR carbamate hydrolase design OR computational design
urethanase RFDiffusion

Searched web

UMG-SP2 urethanase Paiva OR Świderek

10

Searched web

AI generative design segmented polyurethane OR elastane OR spandex
cleavable OR recyclable polymer design polyBART POLYT5 vitrimer

10

Searched web

enzymatic recycling polyurethane OR elastane OR spandex "urethanase" OR
carbamate hydrolase site:arxiv.org OR site:biorxiv.org OR
site:chemrxiv.org

10

Analyzing prior art for urethanases

Searched web

elastane OR spandex enzymatic degradation OR recycling urethanase OR
carbamate hydrolase

10

Searched web

"segmented polyurethane" OR elastane OR spandex "generative" OR "AI
design" OR "machine learning design" OR polyBART OR POLYT5 OR vitrimer

5

Searched web

joint polymer enzyme co-design OR co-optimization OR coupled design
recycling OR circularity

5

Browsed

science.org/doi/10.1126/science.adu2454

Researching technical feasibility of segmented copolymers

Searched web

Boltz-2 protein structure prediction OR PLACER enzyme design

10

Searched web

UMG-SP2 urethanase catalytic triad OR mechanism Ser Ser Lys OR Ser-His-Asp

10

Browsed

pubs.rsc.org/en/content/articlelanding/2025/sc/d4sc06688j

Searched web

elastane OR spandex hard segment crystallinity enzymatic degradation
barrier OR challenge

5

Evaluating enzymatic degradation feasibility

Searched web

Lauko computational design serine hydrolases Science 2025 DOI OR date

10

Searched web

POLYT5 polymer design Georgia Tech elastane OR polyurethane OR segmented
OR block copolymer

10

Writing the final threat assessment report